

**EduRAG: Multimodal Retrieval-Augmented
Generation for Educational Video Understanding**

**A PROJECT REPORT SUBMITTED IN PARTIAL
FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD
OF THE DEGREE OF**

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING**

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**DEPARTMENT OF COMPUTER SCIENCE AND
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March 2026



VNRVJIET

**VALLURUPALLI NAGESWARA RAO
VIGNANA JYOTHI INSTITUTE OF
ENGINEERING & TECHNOLOGY**

AUTONOMOUS INSTITUTION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING ENGINEERING

CERTIFICATE

This is to certify that the Minor Project titled “**TITLE OF THE MINOR PROJECT**” is a record of bona fide work submitted by:

- **NAME OF THE STUDENT 1** **23071Axxxx**
- **NAME OF THE STUDENT 2** **23071Axxxx**
- **NAME OF THE STUDENT 3** **23071Axxxx**
- **NAME OF THE STUDENT 4** **23071Axxxx**

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech.)** in **COMPUTER SCIENCE AND ENGINEERING** of **Jawaharlal Nehru Technological University Hyderabad (JNTUH)**, under **VNRVJIET B.Tech. R22 Academic Regulations**. The work has been carried out under our guidance and supervision at **Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad**.

The report is original, free from plagiarism, and complies with the Institute guidelines, including those related to academic ethics and AI-generated content.

Signature of Supervisor

Name of the Supervisor

Designation of the Supervisor

Xxxx Engineering

Signature of HOD

Dr. XXXXXXXX

Professor & HOD

XXXXXX Engineering

VNR VJIET, Hyderabad

VNR VJIET, Hyderabad

PLAGIARISM CERTIFICATE

APPROVAL CERTIFICATE

Minor Project evaluation for the project titled “**TITLE OF THE MINOR PROJECT**”, being submitted by

Mr./Ms. NAME OF THE STUDENT 1, Regd. No. **23071Axxxx**,

Mr./Ms. NAME OF THE STUDENT 2, Regd. No. **23071Axxxx**,

Mr./Ms. NAME OF THE STUDENT 3, Regd. No. **23071Axxxx**,

Mr./Ms. NAME OF THE STUDENT 4, Regd. No. **23071Axxxx**,

has been conducted on _____ and the work is approved for the award of the degree of **BACHELOR OF TECHNOLOGY (B.Tech.)** in **COMPUTER SCIENCE AND ENGINEERING**, under VNRVJIET B.Tech. R22 Academic Regulations.

INTERNAL EXAMINER

EXTERNAL EXAMINER



VNRVJIET

**VALLURUPALLI NAGESWARA RAO
VIGNANA JYOTHI INSTITUTE OF
ENGINEERING & TECHNOLOGY**

AUTONOMOUS INSTITUTION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We, the undersigned, declare that the Minor Project titled “**TITLE OF THE MINOR PROJECT**” has been carried out by us in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech.)** in **COMPUTER SCIENCE AND ENGINEERING** of Jawaharlal Nehru Technological University Hyderabad (JNTUH), under VNRVJIET B.Tech. R22 Academic Regulations, at Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad.

We further declare that this work is original, has not been submitted previously for any degree or similar title, and complies with the Institute’s plagiarism and academic ethics guidelines, including AI-generated content. The contributions of all team members are clearly indicated in the report.

PLACE: HYDERABAD

DATE: _____

Signature of the Student(s)

NAME OF THE STUDENT 1

23071Axxxx

NAME OF THE STUDENT 2

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ABSTRACT

The report should present the project work in a clear, logical, and well-structured manner.

The abstract is a concise summary of the Minor Project Phase–II report, typically limited to 200–300 words. It briefly presents the background and motivation of the problem, clearly states the problem definition (aligned with Phase–I), outlines the objectives of the project, and summarizes the approach adopted by the team. It also highlights the work carried out during Phase–II, the key results obtained, and the extent to which the objectives have been validated. The abstract should be written as a single paragraph (generally 5–10 sentences) and must provide a clear overview of the entire report without including detailed explanations, figures, or references. It should effectively highlight:

- The relevance of the problem addressed
- The objectives of the project
- The approach or methodology adopted
- The work carried out during Phase–II
- The key results obtained and validation of objectives
- The major conclusions or outcomes

At the end of the abstract, include 3–5 keywords that represent the main aspects of the project and facilitate indexing and search in repositories or digital libraries.

Keywords: Keyword1, Keyword2, Keyword3

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- 4.1 Implementation Plan
- 4.2 Development / Execution of Major Components
- 4.3 Integration and Overall Functioning
- 4.4 Challenges Encountered and Solutions
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- 5.1 Testing / Evaluation Methodology
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- 5.3 Analysis and Interpretation of Results
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CHAPTER 6 – CONCLUSIONS AND FUTURE SCOPE

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REFERENCES

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NOMENCLATURE

List all symbols, abbreviations and notations used in the report.

SG Sample Graph

ACKNOWLEDGEMENT

The Acknowledgement section is used to express gratitude and appreciation to individuals and organizations who have supported the students during the execution of the Minor Project Phase–II work.

Acknowledge the following:

- Project Supervisor for guidance, technical support, and continuous monitoring of the project work
- Head of the Department for providing facilities, encouragement, and administrative support
- Faculty members / staff who contributed academically or technically during the course of the project
- Institution for providing infrastructure, resources, and a conducive environment for carrying out the project work
- Any other individual or organization that has provided significant academic or technical support

NAME OF THE STUDENT 1

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CHAPTER 1

INTRODUCTION

The report should present technical content in a clear, logical, and well-structured manner.

The report shall follow a structured format with proper organization of chapters, headings, figures, tables, and references.

This chapter should clearly introduce the project so that the reader understands what the problem is, why it is important, and what the project aims to achieve.

Include:

- Background of the problem
- Need and relevance of the work
- Problem statement
- Objectives of the project
- Scope of the work carried out
- Organization of the report

CHAPTER 2

LITERATURE REVIEW

This chapter should present a review of existing work related to the project.

- Use recent and relevant sources (journals, conferences, books, patents)
- Organize the literature logically (year-wise, method-wise, or theme-wise)
- Compare different methods or approaches used in similar problems
- Use tables or figures, wherever appropriate, for comparison
- Do not just describe the literature; analyze and compare the work
- Identify limitations, trends, and research gaps
- Clearly establish the need for the present work
- Ensure all sources are properly cited

CHAPTER 3

APPROACH / DESIGN METHODOLOGY

This chapter should present the approach adopted for carrying out the project work and demonstrate the technical reasoning behind the selected methodology.

Include:

- Brief overview of Phase–I work and key outcomes
- Refinement of problem and objectives based on feedback
- Different possible approaches considered for solving the problem
- Criteria used for selecting the final approach
- Description of the selected approach and its key elements
- Supporting analysis, assumptions, and design considerations
- Tools, techniques, or methods used
- Justification of the selected approach with respect to objectives

Architectural Diagram, DFD, Flowcharts, UML diagrams and relevant representations may be included within the description wherever required.

CHAPTER 4

DEVELOPMENT AND IMPLEMENTATION

This chapter should present the work carried out during Phase–II and demonstrate the progress of development and execution of the project.

Include:

- Implementation plan and progress with respect to defined milestones
- Development or execution of major components or stages of work
- Integration of different parts and overall functioning of the work
- Challenges encountered during execution and the solutions adopted
- Key observations during development and implementation

CHAPTER 5

RESULTS, ANALYSIS AND VALIDATION

This chapter should present the results obtained from the project work and demonstrate their analysis and validation with respect to the defined objectives.

Include:

- Method used for testing, evaluation, or verification
- Results obtained from the work carried out
- Analysis and interpretation of results
- Validation of results with respect to project objectives
- Comparison with expected outcomes or existing work (if applicable)

Use tables, graphs, and figures wherever appropriate to present results clearly.

Focus on interpretation and justification rather than only presenting data.

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

This chapter should summarize the overall outcomes of the project and present the scope for further improvement.

Include:

- Key conclusions derived from the work carried out
- Extent to which the project objectives have been achieved
- Limitations of the work
- Possible improvements and future scope

Keep the content concise and focused.

Do not introduce new concepts in this chapter.

REFERENCES

All references must be listed at the end of the report using a consistent citation style. The Institute recommends using the **IEEE citation style**.

Students are encouraged to use reference management tools such as **Mendeley Reference Manager, Zotero, or EndNote** to organize and format citations.

General Instructions:

- Cite all sources used in the project work, including literature review, methodology, and results discussion
- Use numbered references in the order of citation (e.g., [1], [2], [3])
- Ensure every cited reference appears in the reference list
- Do not include sources that are not cited in the report
- Maintain uniform formatting throughout the report
- Prefer recent and credible sources such as journals, conferences, standards, and technical reports

Examples (IEEE Style)

Book

[1] D. Sarunyagate, Ed., *Lasers*. New York: McGraw-Hill, 1996.

Chapter in a Book

[2] G. O. Young, “Synthetic structure of industrial plastics,” in *Plastics*, 2nd ed., vol. 3, J. Peters, Ed. New York: McGraw-Hill, 1964, pp. 15–64.

eBook

[3] L. Bass, P. Clements, and R. Kazman, *Software Architecture in Practice*, 2nd ed. Boston: Addison-Wesley, 2003. [Online]. Available: <https://www.example.com>

Journal Article

[4] G. Liu, K. Y. Lee, and H. F. Jordan, “TDM and TWDM de Bruijn networks and shufflenets for optical communications,” *IEEE Trans. Computers*, vol. 46, no. 6, pp. 695–701, Jun. 1997.

Conference Paper

[5] L. Liu and H. Miao, “A specification-based approach to testing polymorphic attributes,” in *Proc. 6th Int. Conf. Formal Engineering Methods (ICFEM)*, Seattle, WA, USA, Nov. 2004, pp. 306–319.

Conference Proceedings (Edited Book)

[6] T. J. van Weert and R. K. Munro, Eds., *Informatics and the Digital Society*. Boston: Kluwer Academic, 2003.

Technical Report

[7] K. E. Elliott and C. M. Greene, “A local adaptive protocol,” Argonne National Laboratory, Tech. Rep. 916-1010-BB, 1997.

Patent

[8] J. P. Wilkinson, “Nonlinear resonant circuit devices,” U.S. Patent 3,624,125, Jul. 16, 1990.

Standard

[9] *IEEE Criteria for Class IE Electric Systems*, IEEE Standard 308, 1969.

Thesis / Dissertation

[10] J. O. Williams, “Narrow-band analyzer,” Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, 1993.

APPENDICES (IF APPLICABLE)

Include additional supporting material that is relevant to the project work but not included in the main chapters. Such material may include:

- Detailed calculations
- Additional diagrams, drawings, or representations
- Data tables or datasets
- Code snippets / algorithms (if applicable)
- Experimental details or supplementary results

Label each appendix clearly (Appendix A, Appendix B, etc.) and refer to them appropriately in the report wherever required.

APPENDIX A: CONTRIBUTION OF TEAM MEMBERS

This appendix should present the contribution of each team member in carrying out the project work.

Include:

- Name and Registration Number of each team member
- Major tasks or components handled by each member
- Nature of contribution in different stages such as planning, development, testing, and documentation

The contribution should be clearly defined and mutually agreed upon by all team members.